Correct Verdicts, Wrong Field: Attribution and Capability Are Dissociable When Small Language Models Follow a Rubric

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Abstract

A behavioral format assay changes the structure of a prompt and reads whether a model's answers improve. Verdict accuracy is the usual measure. We show that it is not sufficient, and that a model's self-reported basis for its answer is not sufficient either. We gave three local language models a structured decision rubric that names one field, a scope condition, as the field on which the verdict turns, and asked each for a verdict and for the field that determined it. Across six decision classes and two scenario polarities, verdict accuracy sat at or near ceiling (0.96–1.00), while the share of correct verdicts that cited the scope field varied widely by model, from 0.05 to 0.66. On the firing scenarios a 32B model cited the scope field in none of its 48 correct verdicts, naming the Marker recognition axis instead; the thinking model named the scope field in 15 of 48. The non-scope citation is model-specific: a recognition axis for the two Qwen models, the Pull character for Mistral. Yet the cited field is not the reasoning: in all 33 of the thinking model's correct base runs that cited a non-scope field, the recorded reasoning trace worked the scope condition and then named a different field. The self-report is unfaithful. A grader scoring only the verdict, or trusting the cited field, would misread both. We then removed the two recognition axes. Accuracy did not fall, but it was at ceiling in both arms, so the accuracy test of fragility is uninformative; what moved was the citation, toward the scope field (the thinking model to 96 of 96) and, for the 32B model, toward the surviving Rest axis (scope 65, Rest 30 of 96). The named field tracks which fields are present, not a fixed dependency. Attribution is dissociable from capability: the field a model reports is neither the field its reasoning used nor the only field it can name. We argue that a format assay must not read the verdict or the model's self-reported field as evidence that the intended field did the work; the self-report is checked against the trace and against an intervention, not trusted.

1 Introduction

A common way to test whether a change to a prompt's structure helps a language model is to score the model's answers before and after. If accuracy rises, the structure gets the credit. This paper is about a failure of that inference. A model can reach the correct answer through a part of the prompt that the change was not about, so that a rise in accuracy is not evidence that the intended structure did any work.

We study this with a structured decision rubric. The rubric describes a class of decision an agent faces, and it names one field, a scope condition, as the field on which the verdict turns. It also carries two recognition axes that describe what a violation and a correct action look like. The rubric states the order of evaluation: read the scope condition, then render the verdict. We ask a model for a verdict on a concrete scenario and, on the same turn, for the name of the field that determined its verdict. The verdict is scored against ground truth. The named field is scored against the field the rubric designates.

Two questions follow. First, does verdict accuracy track which field the model cites as its basis? We find it does not: accuracy is near ceiling while scope-field citation is far lower, and lowest on the scenarios where the decision class fires. And the cited field is a self-report that is unfaithful to the reasoning: for the thinking model, whose reasoning we record, every correct run that named a non-scope field had in fact worked the scope condition in its trace. Second, does a correct-but-off-field citation depend on the field the model names? We test this by removing the two recognition axes. Accuracy holds, though it was already at ceiling, and the citation shifts toward the scope field and, where it survives, the Rest axis. The field a model names is a preference among the fields present, not a requirement.

We call the resulting distinction *attribution versus capability.* What a model reports as the basis of its answer (attribution) is separable both from whether the answer is correct (verdict accuracy) and from which fields the model can actually use to be correct (capability). A behavioral format assay that reads only the verdict sees none of this. It cannot tell a genuine effect of the intended structure from a rise carried by an unintended field, and it cannot tell a load-bearing field from one the model merely names.

1.1 Related Work

That a model's stated reasoning need not reflect the cause of its answer is established for chain-of-thought prompting. Turpin et al. [2023] show that models give plausible reasoning that does not mention the features actually driving their predictions, and Lanham et al. [2023] measure how far a stated chain of thought is load-bearing for the final answer. Chen et al. [2025] report the same for reasoning models, whose stated chains omit decisive influences such as an injected hint. Our field-source measure is a compact form of the same question for a rubric-following task: the model names the field it used, and we check that name against an intervention rather than taking it at face value. The distinction between an answer's correctness and the process that produced it also motivates process-level supervision, where a reward is placed on the reasoning steps rather than the outcome [Lightman et al., 2023]. Our result is a reason that outcome scoring alone is insufficient for a format study, framed as a measurement problem rather than a training one.

A model reporting an unintended feature as its basis echoes the shortcut-learning pattern [Geirhos et al., 2020], described there for learned representations; we observe it at the level of a model's self-reported field on a structured prompt, and we add two things: the self-report is unfaithful to the reasoning (the model works the scope condition regardless), and it is not load-bearing (its citation shifts when the named field is removed, at no cost to accuracy). The rubric we use comes from a research program on descriptive decision-point prompts [Neilson, in preparation]; two adjacent studies of format-dependent behavioral shifts in language models [Wang et al., 2026, Liu et al., 2026] vary prompt structure and read behavioral output, but neither separates the model's self-reported route from its verdict, which is the measure this paper turns on.

2 Materials and Method

2.1 The Rubric and the Field-Source Measure

Each decision class is described by a rubric with a fixed shape. A **Y − Decision terrain** block names the evaluation order and three parts: a **Pull character** (why the locally-wrong action is attractive, present in both firing and non-firing cases), a **Scope-operative-when** field (the condition under which the decision class fires, with a verification procedure), and a **Scope-not-operative-when** field (the condition under which it does not, stated as a positive discriminator). Below the terrain sit two axes: a **Marker axis** (an invariant describing what the violation looks like) and an **Aim axis** (an invariant describing correct navigation). A **Rest**

axis describes territory where the class does not arise. The terrain block states: “Evaluate in sequence: pull character → scope → verdict. Do not render a verdict before completing the scope check.” The Scope field is therefore the field the rubric designates as the decision route; the Marker and Aim axes are recognition and characterization, not the route.

We use six decision classes (Section B). Each has two scenarios: a *firing* scenario whose ground truth is “yes” (the class applies) and a *carve-out* scenario whose ground truth is “no” (a named exception holds). The intended field for a firing scenario is Scope-operative-when; for a carve-out scenario it is Scope-not-operative-when. Both are the Scope field.

The instrument is the response format. The model is asked to answer in four lines: a `VERDICT` (yes/no), a `FIELD SOURCE` (the name of the field that determined the verdict), an `OBSERVABLE` (the condition it cites), and the `EVIDENCE` (the trace element it rests on). The verdict and the field source are parsed deterministically. A *wrong-path success* is a correct verdict whose cited field is not the Scope field.

2.2 Model and Sampling

Three subjects, all Q4_K_M GGUF quantizations served by llama.cpp (`llama-server`, build `b9445-af6528e6d`) over the raw completion endpoint, so no server-side chat template sits between the declared prompt and the model. Each model’s instruct wrapper is published verbatim (Appendix A).

- **Mistral-7B-Instruct-v0.3** (7.2B), non-thinking.
- **Qwen2.5-32B-Instruct** (32B), non-thinking.
- **Qwen3.8-27B** (27B), thinking enabled. Its reasoning trace is recorded per run.

Claude-family models are excluded as subjects: the rubric corpus is in their training data. The sampler chain is declared complete and identical across subjects: temperature 0.8, `min_p` 0.05 as the sole truncation stage, every other truncation stage and every penalty disabled, per-run seeds 1–8. Penalties are off by design: the rubric induces repeated structure (axis-by-axis traversal), and a repetition penalty would attenuate the behavior under study. The served context was 32768 tokens for Mistral and Qwen3.8. Qwen2.5-32B was served at 8192 tokens: its weights plus the KV cache at 32768 exceed the 24 GB GPU. The prompts are about 3800 tokens with at most 1024 generated, so 8192 has ample headroom and no run truncated; the served context is recorded per run and bounds only the unused tail.

2.3 Conditions and Arms

Every run assembles one prompt: the shared rubric-definition document, then the decision-class terrain, then the scenario trace, then the four-line instruction. The study has two arms that differ only in the terrain:

- **Base arm.** The full terrain, with the Marker and Aim axes present.
- **Ablation arm.** The same terrain with the Marker and Aim axes removed. The Y-Decision block (including both Scope fields), the Pull character, and the Rest axis are byte-identical to the base terrain; a per-file diff confirms only the two axes changed.

The design is 3 subjects × 2 arms × 6 classes × 2 scenarios × 8 runs = 576 runs. No run truncated, and the served model matched the declared model in every run.

2.4 Scoring

The headline measures are deterministic. The verdict is parsed and compared to ground truth. The field source is parsed and classified by the first field-family keyword it names: Scope, Marker, Aim, Pull, Rest, or other. For each cell (subject, arm, class, scenario) we report verdict accuracy over the eight runs, and among the correct verdicts the share that cited the Scope field. A run's answer is the four-line response; for the thinking model the reasoning trace precedes it and is recorded separately, so the four parsed lines are the same shape for all subjects. Read cells and direction, not exact counts: at eight runs a single-integer difference is inside the noise band.

2.5 Agents and Models

Table 1 lists every model and agent with a role in producing the results. The three subjects are the models under test. A Claude session authored the study and the analysis code; it is an agent, not a person, and it judged nothing in the deterministic layer, which is a parser.

Table 1: Agents and models.

Role	Model	Version and runtime	Date
Subject (non-thinking)	Mistral-7B-Instruct-v0.3	Q4_K_M, llama.cpp b9445, ctx 32768	2026-09-11
Subject (non-thinking)	Qwen2.5-32B-Instruct	Q4_K_M, llama.cpp b9445, ctx 8192	2026-09-11
Subject (thinking)	Qwen3.8-27B	Q4_K_M, llama.cpp b9445, ctx 32768, thinking on	2026-09-11
Study and analysis author	Claude (agent)	Version not recorded	2026-09-11

3 Results

3.1 The Cited Field Is Usually Not the Scope Field

Verdict accuracy is at or near ceiling for every subject, while the share of correct verdicts that cite the intended Scope field is far lower (Table 2). The models are right, and they largely say a different field made them right.

Table 2: Base arm, per subject, over 96 runs each (12 cells $\times$ 8).

Subject	Verdict accuracy	Scope-cited / correct	Non-scope / correct
Mistral-7B	0.96 (92/96)	0.66	0.34
Qwen2.5-32B	1.00 (96/96)	0.04	0.96
Qwen3.8	1.00 (96/96)	0.66	0.34

Qwen2.5-32B is the sharpest case. It answered correctly in all 96 runs and named the Marker axis in 88 of them, the scope field in four. A grader scoring only its verdicts would read a perfect score and credit the scope structure for a result the model attributes almost entirely to a recognition axis.

The gap is widest on the firing scenarios. Pooling the base arm across subjects, verdict accuracy is 0.99 on both polarities (142/144 each), but scope-field citation among correct verdicts is 0.27 on the firing scenarios against 0.63 on the carve-outs. The non-scope citation is model-specific. On firing scenarios the two Qwen models name the Marker axis's firing condition

(Qwen2.5-32B in all 48 of its correct firing runs, Qwen3.8 in 32 of 48), while Mistral most often names the Pull character (14 of its 46 correct firing runs, against 8 for the Marker axis and 23 for the scope field). The field a correct verdict is attributed to is, in general, not the scope field the rubric puts on the decision path.

3.2 The Cited Field Is Not the Reasoning

For Qwen3.8 the reasoning trace is on the record, so the cited field can be checked against the reasoning it summarizes. In the base arm, 33 of Qwen3.8's 96 correct runs cited a non-scope field (32 the Marker axis, one other). In every one of those 33, the trace worked the scope condition before naming a different field. A representative run (cas-a, run 4) states in its trace "So scope IS met. The glyph may fire.", then deliberates and concludes "The scope check is necessary but it's a gate. The actual firing determination comes from the Marker axis firing condition.", and cites the Marker axis. The scope field is engaged; the citation names an axis. The self-report is not a readout of the reasoning. Only the thinking model exposes this directly, because only its reasoning is recorded; the non-thinking models' citations may conceal the same gap.

3.3 Removing the Shortcut Does Not Break the Answer

If a correct verdict depended on the recognition axis the model cited, removing that axis should collapse the cell. It does not. In the ablation arm, with the Marker and Aim axes deleted, verdict accuracy holds at ceiling for all three subjects. The pre-registered test of fragility was an accuracy differential between the base arm's shortcut-citing and scope-citing cells; with accuracy at ceiling in both arms there is almost nothing to differentiate, so that test is uninformative rather than a clean demonstration of robustness. For Qwen2.5-32B only one of twelve base cells was scope-citing, which compounds the problem. The only accuracy movements are two small *improvements* under ablation for Mistral (one carve-out cell 6/8 → 8/8, one firing cell 6/8 → 7/8).

What moves is the cited field (Table 3). With the recognition axes gone, the citation shifts toward the two fields the ablation leaves in place, the scope condition and the Rest axis, at no cost to accuracy. Qwen3.8 cites the scope field in all 96 ablation runs, up from 63. Qwen2.5-32B, which had cited the Marker axis 88 times, now splits between the scope field (65) and the Rest axis (30); the Rest axis was not ablated, so it remains available as a field to name. Mistral shifts from 63 to 73 scope citations, with most of the remainder on the Pull character.

Table 3: Field-source class over all 96 runs per subject per arm. Verdict accuracy is at ceiling in both arms. The ablation removes the Marker and Aim axes; the two Scope fields, the Pull character, and the Rest axis remain, so a citation can still land on them.

Subject	Base arm	Ablation arm
Mistral-7B	scope 63, pull 15, marker 9, aim 1, rest 7, other 1	scope 73, pull 21, rest 2
Qwen2.5-32B	marker 88, scope 4, rest 4	scope 65, rest 30, other 1
Qwen3.8	marker 32, scope 63, other 1	scope 96

The scope field was available throughout. The models named the recognition axis when it was present and named the scope field, or the Rest axis, when it was not, at no change in accuracy. The named field is a preference among the fields present, not a dependency.

4 Discussion

4.1 Attribution Is Not Capability

Several quantities that a verdict-only assay silently conflates come apart here. The *verdict* is whether the answer is correct. The *attribution* is the field the model names as its basis. The *capability* is which fields the model can use and still be correct. A fourth, visible only for the thinking model, is the *reasoning*: the fields the trace actually works. Verdict accuracy is near ceiling regardless of the rest. Attribution is mostly off the scope field in the base arm. The reasoning engages the scope field even when the attribution does not, in all 33 non-scope citations we could check. Capability includes the scope field throughout, as the ablation shows. Knowing the verdict tells you nothing about the attribution; knowing the attribution tells you nothing about the reasoning or the capability.

For a behavioral format assay this is a direct constraint. A rise in verdict accuracy after a structural change is not evidence that the intended structure did the work: the model may attribute the answer to another field, and may reason through the intended field while naming another. A field the model never names may still be one it uses and one it can fall back on, so the absence of a citation is not evidence of an absent capability. The assay must not read either the verdict or the model's self-reported field as evidence that the intended field carried the result.

4.2 The Self-Report Must Be Checked, Not Trusted

The field-source line is the model's account of its own decision. Taken at face value it would say that Qwen2.5-32B is a Marker-axis reasoner that ignores the scope condition. Two controls refute that. The reasoning trace, where we have it, shows the scope condition worked before a non-scope field is named. And the ablation: remove the Marker and Aim axes and the same model names the scope field (65 of 96) and the Rest axis (30), and stays correct. The self-report described a preference, not a dependency, and not the reasoning. This is the chain-of-thought faithfulness problem [Turpin et al., 2023, Lanham et al., 2023, Chen et al., 2025] on a rubric-following task, with two cheap controls: read the recorded trace, and delete the named field and re-measure. We recommend both for any assay that reads a model's stated basis.

4.3 Relation to the Orientation Study

An earlier small study (four runs per cell, on a since-retired inference runtime with an unrecorded sampler) had found that adding structure to one decision class drove its firing scenario from three-of-four correct to zero, and read this as evidence of no reliable correct path beneath a shortcut. The present study, run clean and at larger scale, does not support that reading. The earlier intervention *added* a routing block; ours *removes* the shortcut. When the shortcut is removed, the answer survives. We take the earlier collapse to have been an effect of the added structure, not evidence of a missing path, and we report the correction here because the orientation study's framing motivated this one.

4.4 Limitations

Eight runs per cell support statements about direction and about large gaps, not about small differences between counts. We ran no no-terrain or scrambled baseline, so we do not claim the terrain is necessary for the verdict; near-ceiling accuracy may mean the scenarios are answerable without it. The claim is about which field a correct verdict is attributed to given the terrain, not about whether the terrain produced the verdict. We removed the Marker and Aim axes but not the Scope fields or the Rest axis, so we do not claim the scope field is uniquely the fallback; the ablation shows only that the intended field and the Rest axis are available and named. The field-source classifier keys on the first field-family term a response names; the Rest axis, which the ablation leaves in place, absorbs 30 of Qwen2.5-32B's ablation citations, and one response still resolves only to "other". The observable-and-evidence quality of each response (whether the cited condition is the discriminating one) is a separate, judge-scored layer we do not report here. The reused `cas` decision class is a known compound entry flagged in the corpus record for terrain-format issues; we reused it verbatim from the orientation study for continuity, not as a clean exemplar. The materials call the rubric a "glyph specification", the term from the research program it comes from; we use "rubric" and "decision terrain" for a general reader. The three subjects are small open-weight models on one GPU; we do not claim the exact rates transfer to larger or hosted models, only that the dissociation appears across two families, two scales, and both reasoning modes. Qwen2.5-32B ran at a smaller served context than the other two subjects; the prompts fit within it, so this bounds only unused context, but it is an inconsistency in the rig, recorded rather than hidden.

5 Conclusion

A structured decision rubric that names its decision field lets us separate what a verdict-only assay treats as one: whether a model is right, which field it says made it right, which fields it can use to be right, and, for a thinking model, which field its reasoning works. Across three small language models the verdict was near ceiling, the cited field was mostly off the scope field, and the scope field was usable throughout. Verdict accuracy did not track the cited field, and the cited field did not track the reasoning: for the thinking model, every correct non-scope citation we checked had worked the scope condition in its trace. The self-report described a preference that an ablation showed was not a dependency. A behavioral format assay should therefore not read the verdict, or the model's self-reported field, as evidence that the intended field carried the result; the self-report is checked against the trace and against an intervention. Attribution, reasoning, and capability come apart, and none is read off the verdict.

5.1 Data Availability

The complete study (rubric-definition document, the six decision-class terrains in both the full and ablated forms, the twelve scenarios, the four-line instruction, the generated study definitions, every run's rendered prompt, response, and (for the thinking model) reasoning trace, the scored grids, and the scoring script) is available at `https://github.com/sophdn/corpos-lab` under the `studies/wrong-path-field-source/` directory.

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A Materials

The instruction, given to every subject after the assembled material, is reproduced verbatim:

> Read the glyph and the trace above.
>
> Respond using exactly this format — one line per field, no additional text:
>
> VERDICT: yes or no
> FIELD SOURCE: the name of the specific field in the glyph specification that determined your verdict
> OBSERVABLE: quote or closely paraphrase the specific condition from that field
> EVIDENCE: state which element of the trace your verdict rests on

The per-subject instruct wrappers, into which the assembled material is substituted, are: `[INST] {prompt} [/INST]` for Mistral; `<|im_start|>user\n{prompt}<|im_end|>\n<|im_start|>assistan` for both Qwen subjects, with no pre-filled think block so Qwen3.8 reasons.

B Decision Classes

The six decision classes are `cas` (companion-artifact-scope-gap), `cgu` (conditional-gate-uniform-default), `fsb` (formal-step-context-bypass), `gop` (governed-operation-protocol-bypass), `psc` (parent-state-check-bypass), and `scb` (structural-ceiling-bypass). Each terrain, in full and ablated form, and each scenario, is in the study directory named in Data Availability.